INCH-POUND

MIL-I-14729A (AR)
AMENDMENT 3
19 May 1994
SUPERSEDING
AMENDMENT 2
27 August 1990

MILITARY SPECIFICATION

INCREMENT, PROPELLANT, M90A1

This Amendment forms a part of Military Specification MIL-I-14729A (MU), dated 5 June 1969, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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Title, Delete "M90E1" wherever it appears and substitute "M90A1."

* 2.1, Under SPECIFICATIONS, MILITARY, delete "MIL-B-117, MIL-A-2550 and MIL-I-45607" and replace with:

"MIL-B-117 - Bags, Sleeves and Tubing
MIL-A-2550 - Ammunition, Acquisition, General Specification for
MIL-I-45607 - Inspection Equipment, Maintenance and Disposition of"

Add "MIL-F-50449 - Filler, Sheet Form"

Under STANDARDS, MILITARY, add:

"MIL-STD-129-1 - Marking for Shipment and Storage-Ammunition and Explosives
MIL-STD-1167 - Ammunition Data Card"

AMSC N/A FSC 1375
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* 2.1, Under DRAWING, ARMY, delete "20-4-619, 7548645, 8858577 and 8858848."

* 2.2, CODE OF FEDERAL REGULATIONS, change from "49 CFR 71-90 - Interstate Commerce Commission Rules and Regulation for Transportation of Explosive and Other Dangerous Articles" to "Title 49, Transportation Parts 100 - 199."

* Delete "(Interstate Commerce Commission are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from Superintendent of Documents, Government Printing Office, Washington 25, D.C.) Orders for the above publication should cite "49 CFR 71-90 (latest revision)."

3.3, Delete "Strength of heat seal" in its entirety.

3.4, Delete "Gross weight of propellant increment" in its entirety and substitute with the following:

"3.4 Gross weight of propellant increment. The weight of the loaded projectile increment within each lot shall vary not more than plus three or minus three grains from the established increment weight (assessed propellant weight plus propellant bag tare weight) (see para. 4.2.3.2)."

Add the following paragraphs:

"3.6 Weight of propellant. The weight of the propellant, loaded into the propellant bag, shall vary not more than plus three or minus three grains from the assessed propellant weight.

3.7 Vacuum test. The loaded, sewn and sealed propellant increments shall withstand a surrounding absolute pressure of 2 to 7 inches of mercury (i.e., 23 to 28 inches of mercury gage vacuum) for 5 seconds minimum without deflating. Increment shall be tested as specified in 4.3.2."

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4.2.2.1, Delete "Critical Defect 1. Bag or bag seam ... propellant can escape" and substitute "Critical Defect 1. Bag or bag seam damaged or stitches missing to extent that propellant can escape."

Delete Major Defects 101 and 102.

Delete Minor Defect "205. Stitches broken or missing" and substitute "205. Stitches broken or unraveled along heat seal (broken stitch permitted at intersection of stitch and heat seal)."

Add Minor Defect "208. End folds not fully sewn (by transverse stitching) ...01010."

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* 4.2.2.2, Delete "or 5.1.3."

* 4.2.3, Delete "or 5.1.3."

4.2.3, Delete 4.2.3.1 and 4.2.3.2 in their entirety and substitute the following:

"4.2.3.1 Weight of propellant. (see 3.6) Critical Defect (under minimum weight); Major Defect (over maximum weight)-Code No. 05001.-Prior to loading, the propellant shall be weighed 100 percent.

4.2.3.2 Gross weight of propellant increment. (see 3.4)-Critical Defect (under minimum weight); Major Defect (over maximum weight)-Code NO. 04001.-The loaded propellant increments shall be weighed 100 percent. Any propellant increment which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4."

4.2.3.3, Delete in its entirety.

Add the following paragraph:

"4.2.3.5 Vacuum test. Critical Defect-Code No. 09001-The propellant increments shall be tested 100 percent. If any increment fails to meet the requirement (see 3.7), it shall be rejected and removed from the lot. Testing shall be performed as specified in 4.3.2 using equipment in accordance with 4.2.4."

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4.3.1.1, Delete in its entirety.

Add the following paragraph:

"4.3.2 Vacuum test. Propellant increments shall be confined to slots 1 1/2 inches deep x 1/2 inch high x a minimum of 6 inches wide for B charges and a minimum of 9 inches wide for A charges. Units of slots may be integrated to comprise an increment rack. Increments, contained in the slots, shall be placed horizontally into a vacuum chamber. Increments shall be subjected to a surrounding absolute pressure of 2 to 7 inches on mercury (i.e., 28 to 23 inches of mercury vacuum). Vacuum shall be drawn gradually and uniformly with respect to time. Vacuum shall be attained in no less than 10 seconds. The increment rack shall be rotated 90 degrees (under vacuum) to enable those defective (non-inflated) increments to fall free of their slots. The rack shall be held in this position for a minimum of 5 seconds. The increment rack shall then be rotated back to its original position. The rack will then contain acceptable increments only. Defective bags (leakers) will have dropped to the bottom of the vacuum chamber.

The rack shall be subjected to an initial and a continuous, Picatinny Arsenal approved, qualification procedure. This procedure shall consist of dumping the loaded rack a number of times without vacuum, and in accordance with the same dumping cycling time used for actual testing. Failure to release increments shall be reason for rack disqualification."

* 5.1.1, Delete paragraph in its entirety and substitute:

"5.1.1 Level A. Propellant increments shall be packed in heat sealed barrier bags complying with Type I, Class F, of MIL-B-117. The barrier bag size shall be 22 1/2 inches long by 22 1/2 inches wide, nominal. Approximately 15 pounds of propellant shall be packed in each bag and closed by means of heat sealing. Two sealed bags shall be packed in a wooden box (Suggested inside box dimensions 14 1/2 inches by 7 1/4 inches.) conforming to Grade A, Type I, Class 1 or 3 of MIL-B-2427. Both ends, sides, top, and bottom of the wooden box shall be lined with 1/16 inch thick solid water resistance sheet material conforming to MIL-F-50499. The box shall be closed and strapped in accordance with MIL-B-2427. The net weight shall not exceed 30 pounds of propellant."

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* 5.1.1.1, Delete paragraph in its entirety.

* 5.1.2, Delete paragraph in its entirety and substitute:

"5.1.2 Level B. The Level B packaging for Propellant Increments shall be same as Level A, 5.1.1."

* 5.1.2.1, Delete paragraph in its entirety.

5.1.3, Delete paragraph in its entirety.

5.1.3.1, Delete paragraph in its entirety.

5.2.1, Delete paragraph in its entirety and substitute:

* "5.2.1 Level A. The nailed wood boxes shall be marked as required in accordance with Drawing 8796522 and the information specified on 5.2.1.1, 5.2.1.1.1 and 5.2.1.2.

* 5.2.1.1 Propelling increments, charge A. The following information shall apply to the boxes:

NSN and DODIC: 1315-00-370-3548-C021
LOT NO: [ ]
NOMENCLATURE: CHARGE, PROPELLING, INCREMENTS A
PROPER SHIPPING NAME: CHARGE, PROPELLING, FOR CANNON
IDENTIFICATION NUMBER: UN0279
GROSS WEIGHT:
NET WEIGHT:
POP MARKING: As specified on 5.2.1.1.1

* 5.2.1.1.1 Wood boxes POP marking. Prior to shipping Propelling Increments, Charge A or B, the containers must successfully pass a complete POP test requirement as specified in Code of Federal Regulations (CFR) title 49. Testing must be performed by a certified POP test laboratory and the results must be documented in a POP test report to be submitted to the Government. Testing may be performed using actual Propelling Increments or a simulant (the testing weight shall be increased by 10%). The POP marking to be applied to the exterior pack shall be as specified by the Government after review and acceptance of submitted POP test report. The marking shall be in accordance with CFR 49 and MIL-STD-129-1. The POP tests

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shall be performed again whenever a change in the packaging or packing manufacture takes place. Testing must be repeated if two years have elapsed since the last test has taken place. The onus of the package POP certification compliance lies with the outer container manufactures.

* 5.2.1.2 Propelling increments, Charge B. The following information shall be applied to the boxes:

NSN and DODIC: 1315-00-378-9841-C022
LOT NO: [_ _ _ _ _ _ _ _ _ _]
NOMENCLATURE: CHARGE, PROPELLING, INCREMENTS B
PROPER SHIPPING NAME: CHARGE, PROPELLING, FOR CANNON
IDENTIFICATION NUMBER: UN0279
GROSS WEIGHT:
NET WEIGHT:
POP MARKING: As specified on 5.2.1.1.1"

* 5.2.2, Delete paragraph in its entirety and substitute the following:

"5.2.2 Level B. The Level B marking for Propellant Increments shall be same as Level A, 5.2.1."

* 5.2.3, Delete paragraph in its entirety.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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(Project 1375-A514)

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